how do I set up Visual Studio Code so I can debug a python script running in a container?

February 02, 2026

To set up Visual Studio Code for debugging a Python script in a container, you can either **attach to a running container** or use **remote debugging** via the `debugpy` library.

Method 1: Attach to a Running Container (Recommended)

This method connects VS Code directly to a container, allowing you to edit and debug code as if it were local.

1. **Install Extensions**: Install the Dev Containers extension and the Python extension on your host machine.
2. **Attach to Container**:
   1. Start your container as you normally would.
   2. Open the **Command Palette** (`Ctrl+Shift+P`) and select **Dev Containers: Attach to Running Container...**.
   3. Select your container from the list. A new VS Code window will open "inside" the container.
3. **Install Inside Container**: Ensure the **Python** and **Python Debugger** extensions are installed in this *new* window (they must be installed in the container environment, not just on your host).
4. **Debug**: Open your Python script, set breakpoints, and press **F5**.

Method 2: Remote Debugging (Attach to Process)

Use this if you want to keep VS Code running on your host machine and only

Ask anything

Use code with caution.

3. Ensure your `docker run` command maps the port: `-p 5678:5678`.

2. **Configure VS Code**:
   1. Create a `.vscode/launch.json` file in your workspace.
   2. Add a "Python Debugger: Remote Attach" configuration:

```json
{
  "name": "Python: Remote Attach",
  "type": "debugpy",
  "request": "attach",
  "connect": { "host": "localhost", "port": 5678 },
  "pathMappings": [{
      "localRoot": "${workspaceFolder}",
      "remoteRoot": "/app" // Path inside your container
  }]
}
```

Use code with caution.

3. **Start Debugging**: Go to the **Run and Debug** view, select your configuration, and click the "Play" button.

Would you like a sample `Dockerfile` that includes these debugging tools by default?

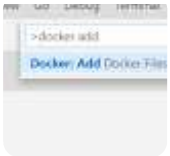
AI responses may include mistakes. Learn more

AI Mode All Images Videos News More

Python in a container - Visual Studio Code

Build, run, and debug the container. The Containers: Add Docker Files to Workspace... command automatically creates a Docker launc...

Visual Studio Code



Debugging Python in Docker: A Tutorial for Beginners

pip install debugpy installs Microsoft's debugpy library. EXPOSE 5678 tells Docker that our container will use port 5678. The CMD ...

KDnuggets

Show all